De La Salle University - Dasmariñas
College of Information and ComputerStudies

COURSE SYLLABUS
Second Semester, SY 2023-2024

COURSE CODE:	**S-ITPE002LA**
COURSE TITLE:	**It Professional Elective 2**
COURSE TYPE:	**Laboratory**
COURSE CREDIT:	2.25 **units**
LEARNING MODALIT:	**Traditional Blended modality**
PRE-REQUISITES:	**None**
CO-REQUISITES:	**S-ITPE002Lec**
PROFESSOR:	**Maria Patricia B. Lina**
CONSULTATION HOURS:	**pblina@dlsud.edu.ph;**
	Thursday 16:00 – 18:00Online/MS Team
	Tuesday 1600-1700 -COS100C

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course introduced the Linux and Open-Source industry and knowledge of the most popular Open-Source applications. The course aims to cover major components of the Linux operating system and explain the technical proficiency to work on the Linux command line. The course has a basic understanding of security and administration related topics server management, working on the command line and permissions.

COURSE LEARNING OUTCOMES:

By the end of the semester, the students can:
CLO1 Identify the system concept, organization, operation. The Operating system evolution, security and process.
CLO2. Apply various types of users on a Linux system, its security and process.
CLO3. Develop students' skills and knowledge in operating system implementation, table structure and device input output system
CLO4. Apply storage management, file, directories and solve storage management problem
CLO5. Developing student critical thinking through the creation of a design paper for a proposed server solution
CLO6. Apply the skills in implementing different kinds of OS especially Linux, and its concept and properties.

LEARNING PLAN:

Course Learning Outcomes	Topic Learning Outcomes
CLO1 Identify the system concept, organization, operation. The Operating system evolution, security and process.	TLO1. Explain what Operating system like Linux is TLO2. Discuss the concepts in (operating) systems and OS major functions.
CLO2. Apply various types of users on a Linux system, its security and process.	TLO3. Discuss the evolution of operating system, its protection to users, to program, and its standard security. TLO4. Discuss different kinds of Linux operation TLO5. Explain the different Linux process
CLO3. Develop students' skills and knowledge in operating system implementation, table structure and device input output system	TLO6. Discuss different kinds of Linux users and their process. TLO7. Explain the system and process security of Linux operating system. TLO8. Explain hardware required for memory protection and the concepts of logical and physical addresses
CLO4. Apply storage management, file, directories and solve storage management problem	TLO9. Discuss the problem of address binding, concept of segmentation and fragmentation. TLO10. Explain what is paging, irs implementation
CLO5. Developing student critical thinking through the creation of a design paper for a proposed server solution	TLO11. Discuss different kinds of table structured and paging interact with segmentation. TLO12. Described benefits of virtual memory system. TLO13. Explain the concepts of demand paging, the working set model and memory replacement and allocation.
CLO6. Apply the skills in in implementing different kinds of OS, and its concept and properties	TLO14. Explain the general structure of the I/O subsystem and the different ways of performing I/O including polling, interruptions, and direct memory access. TLO15. Explain the nature of mass storage, disk storage management, concepts of files, directories and directory.

Midterm Period:

Module No and Learning Outcome(s)	Week Date/s	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Allotted Hours
Module 0,1 CLO 1 CLO 6 TLO 1 TLO 2 TLO 3	Week 1 Aug 19 - Aug 24	• Gospel Reading, Reflection and Prayer • Module 0 - Introduction to the course • Module 1 Getting started Introduction to Operating • Using the shell Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
Module 1 ,2 CLO1 CLO 6 TLO1 TLO 2 TLO 3	Week 2 Aug 26 - Aug 31	• Gospel Reading, Reflection and Prayer • Module 1 Introduction to OS • Configuration and Shell Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
Module 3 CLO 1 CLO 2 CLO 6 TLO 1 TLO3	Week 3 Sept 2 - Sept 7	• Gospel, Reflection and Prayer • Module 2 – Basic file management • File Globing • File manipulation Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
	Week 4 Sept 9 - Sept 14	• Gospel, Reflection and Prayer • Module 3 – Working with text • Text Utilities • Regular Expression Online Enabling Assessment/ Activities	Schoolbook PowerPoint	
Module 4 CLO 2	Week 5 Sept 16	Gospel Reading Reflection and Prayer	Schoolbook PowerPoint	3

CLO 3 CLO 5 TLO 4 TLO 5 TLO 6 TLO 7	- Sept 21	Module 3 – – Working with text • Standard text stream and Redirection • The VI Editor Online Enabling Assessment/ Activities		
Module 5 CLO 2 CLO 3 CLO 5 CLO 6	Week 6 Sept 23 - Sept 28	Gospel, Reflection and Prayer Module 4 – Advance File management • Managing Processes • Active Command Online Enabling Assessment/ Activities	Schoolbook PowerPoint,	3
Module 5A CLO 2 CLO 3 CLO 5	Week 7 Sept 30 - Oct 5	Gospel, Reflection and Prayer Module 5 – – Advance File management • File permission • File system Link Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3
CLO 2 CLO 3 CLO 5 TLO 4 TLO 5 TLO 6 TLO 7 CLO8	Week 8 Oct 7 - Oct 12	Gospel, Reflection and Prayer Module 5 – – Booting system • Hardware configurations • Boot Process Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3
Module 1,2,3, 4,5	Week 9 Oct 14 - Oct 19	Summative assessment: Module 1,2,3,4,5	Schoolbook	
			TOTAL	27

Final Period:

Course Learning Outcomes	**Topic Learning Outcomes**
CLO3. Develop students' skills and knowledge in operating system implementation, table structure and the input output system and devices.	TLO9. Discuss the problem of address binding, concept of segmentation and fragmentation. TLO10. Explain what is paging, its implementation
CLO4. Apply storage management, file, directories and solve storage management problem	TLO11. Discuss different kinds of table structure and paging interact with segmentation. TLO12. Described benefits of virtual memory system.

CLO5. Developing student critical thinking through the creation of a design paper for a proposed server solution	TLO13. Explain the concepts of demand paging, the working set model and memory replacement and allocation. TLO14. Explain the general structure of the I/O subsystem and the different ways of performing I/O including polling, interruptions, and direct memory access.
CLO6. Apply the skills in in implementing different kinds of OS, and its concept and properties	TLO15. Explain the nature of mass storage, disk storage management, concepts of files, directories and directory

Learning Outcomes for Final Period

Module Title and Learning Outcome(s)	**Wek/ Date(s)**	**Teaching-Learning Activities / Assessment Strategy**	**Technology Enabler**	**Allotted Hours** Allotted Hours
CLO 2 CLO 3 CLO 5 TLO 5 TLO 6 TLO 7 CLO8 CLO9	Week10 Oct 21 - Oct 26	Gospel Reading, Reflection and Prayer Module 6 – Booting system •Boot Loader • Run Levels Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
CLO 2 CLO 3 CLO 5 TLO 8 TLO 9 TLO 10 TLO 11	Week11 Oct 28 - Oct 30	Gospel Reading, Reflection and Prayer Module 7 – Partitioning • Designing a Scheme • Creating partition Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
CLO 3 CLO 5 TLO 8 TLO 9 TLO 10 TLO 11	Week 12 Nov 4 - Nov 9	Gospel Reading, Reflection and Prayer Module 8 – Partitioning • Fixing File system • Mounting File system • Maintaining Integrity Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0

CLO 3 CLO 5 CLO 6 TLO 9 TLO 10 TLO 11 TLO12	Week 13 Nov 11 - Nov 16	Gospel Reading, Reflection and Prayer Module 8 – System Software • Maintaining Integrity Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
Module 7 CLO 3 CLO 5 CLO 6 TLO 9 TLO 10 TLO 11 TLO12	Week 14 Nov 18- Nov 23	Gospel Reading, Reflection and Prayer Module 8 – System Software • Package Management • Managing shared Libraries Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
CLO 5 CLO 6 TLO 9 TLO 10 TLO 11 TLO12 TLO13	Week 15 Apr 22- Apr 30	Gospel Reading, Reflection and Prayer Module 8 – System Software • Virtualization Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
Module 8 CLO 3 CLO 4 CLO 5 CLO 6 TLO 11 TLO12 TLO13 TLO14	Week 16 Dec 2- Dec 7	Gospel Reading, Reflection and Prayer Module 10 – Memory Management • Memory Management • Memory information Online Enabling Assessment/ Activities	Schoolbook PowerPoint	3.0
	Week 18 Dec 12- DEc 18	Summative assessment Moduel 6,7,8,9,10	Schoolbook	3.0
			TOTAL	21

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments (Activities, Quizzes, Presentations, etc.)	40%	Enabling Assessments (Activities, Quizzes, Presentations, etc.)	40%
Summative Assessment	60%	Summative Assessment	60%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
    a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
    a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies

a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

Call Number	Book title
HF5415.2 .C43 201	Pelz O. 2018. Fundamentals of Linux : Explore the Essentials of the Linux Command Line. Birmingham: Packt Publishing Ltd.
QA 76.76.O63 .Y165 2014	Yang L. 2018. The Art of Linux Kernel Design. CRC Press.
TK 5105.52 .T599 2016	Tomsho G. 2011. Guide to networking essentials. Clifton Park, N.Y: Delmar ; Andover.
QA 76.76.O63 .C784 2001	DK Publishing, Cooper B. 2001. Linux. DK Publishing (Dorling Kindersley).

Online

Course tutorial. [accessed 2024 August 4. https://lms.netacad.com/course/view.php?id=2278863.

Tutorial: Operating system, Simply Easy Learning. https://www.tutorialspoint.com/operating_system/index.htm

Tutorial: Operating system Tutorial. [accessed 2024 August 4

https://www.geeksforgeeks.org/operating-systems/.

Tutorial: Operating system Process scheduling.

[accessed 2024 August 4 https://www.tutorialspoint.com/operating_system/os_process_scheduling.htm

Tutorial: Operating system Process.

[accessed 2024 August 4] https://www.tutorialspoint.com/operating_system/os_process_scheduling.htm

Tutorial: Operating system Memory management. [accessed 4 August 2024]

https://www.geeksforgeeks.org/memory-management-in-operating-system/

CISCO Networking Academy. [accessed 2024 August 4]. https://www.nettacad.com

EBSCOHOST (Journals)

da G, de J. 2023. NTFS FILE SYSTEM: AN INTRODUCTORY ANALYSIS. Journal of Engineering Research. 3(15):2–6. doi:https://doi.org/10.22533/at.ed.3173152308052.

Silberschatz A, Galvin PB, Gagne G. 2013. Operating System Concepts Essentials, 2nd Edition. Wiley Global Education.

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1st Semester 2024-2025

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